

CHULYM, Russian Federation

WMO: 296250

Lat: 55.100N

Lon: 80.967E

Elev: 142

StdP: 99.64

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -33.0	(d) -39.3	(e) 0.1	(f) -36.4	(g) -35.9	(h) 0.1	(i) -33.0	(j) 10.4	(k) -7.6	(l) 9.2	(m) -8.3	(n) 1.3	(o) 200	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.6	18.8	27.2	18.4	25.6	17.8	20.7	25.9	19.7	24.9	18.8	23.8	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	23.0	17.8	13.0	22.2	16.8	12.2	21.3	60.5	25.9	57.4	25.0	54.2	23.8	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.9	(c) 7.0	(e) -39.3	(f) 31.3	(g) 4.1	(h) 1.7	(i) -42.2	(j) 32.6	(k) -44.6	(l) 33.6	(m) -46.9	(n) 34.6	(o) -49.8	(p) 35.8		
(a) 9.0	(b) 7.9	(c) 7.0	(e) -39.4	(f) 22.4	(g) 3.9	(h) 1.3	(i) -42.2	(j) 23.3	(k) -44.5	(l) 24.0	(m) -46.7	(n) 24.7	(o) -49.6	(p) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-18.5	-15.4	-7.2	3.4	11.5	16.9	18.5	15.8	9.8	2.7	-6.8	-14.7		
	DBStd	14.25	8.55	7.24	6.40	5.79	5.45	4.14	3.21	3.69	4.60	5.10	7.35	8.33		
	HDD10.0	3954	883	712	534	210	49	4	0	2	59	232	504	764		
	HDD18.3	6274	1142	945	793	449	218	73	38	96	258	485	754	1023		
	CDD10.0	822	0	0	0	11	97	210	263	182	53	5	0	0		
	CDD18.3	100	0	0	0	0	7	30	43	17	2	0	0	0		
	CDH23.3	957	0	0	0	3	113	294	342	171	35	0	0	0		
	CDH26.7	184	0	0	0	0	25	56	68	29	7	0	0	0		
(14) Wind	WSAvg	3.2	3.3	3.4	3.6	3.6	3.6	2.8	2.5	2.7	2.9	3.3	3.6	3.8		
(15) Precipitation	PrecAvg	381	17	11	11	20	34	52	58	57	36	34	28	18		
	PrecMax	609	46	28	24	51	94	215	153	104	83	84	60	40		
	PrecMin	240	5	2	0	1	10	9	9	16	2	11	2	2		
	PrecStd	75	10	6	6	11	18	39	36	24	20	17	15	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.0	7.0	22.6	29.3	30.1	30.8	29.4	27.4	19.4	8.0	1.5		
		MCWB	-0.2	0.0	3.0	11.7	16.4	18.9	20.7	19.3	16.6	11.1	6.3	0.2		
	2%	DB	-2.0	-0.9	4.2	18.5	26.3	28.3	28.6	27.1	23.5	15.8	5.6	0.3		
		MCWB	-2.7	-1.8	1.5	10.6	15.1	18.4	19.9	19.2	14.8	9.5	3.8	-0.4		
	5%	DB	-3.9	-2.9	2.7	15.4	23.9	26.7	26.9	25.2	20.5	12.6	3.7	-1.6		
		MCWB	-4.6	-3.8	0.8	9.1	14.0	17.9	19.3	17.9	13.5	8.1	2.3	-2.3		
	10%	DB	-5.8	-4.8	1.3	12.4	21.3	24.9	25.2	23.2	18.0	9.9	1.6	-3.6		
		MCWB	-6.5	-5.6	-0.3	7.5	12.8	17.1	18.8	16.8	12.4	6.7	0.5	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.1	3.4	13.3	17.9	20.8	22.3	21.3	17.7	11.8	6.7	0.6		
		MCDB	0.5	0.7	6.1	20.1	26.9	27.4	28.0	25.9	23.0	16.8	7.7	1.2		
	2%	WB	-2.6	-1.7	1.9	11.3	16.2	19.7	21.1	19.8	15.6	10.1	4.1	-0.3		
		MCDB	-2.0	-0.9	3.6	17.5	24.4	26.0	26.4	25.2	22.2	14.5	5.5	0.3		
	5%	WB	-4.6	-3.7	1.0	9.4	14.8	18.7	20.3	18.7	14.1	8.5	2.3	-2.3		
		MCDB	-3.9	-2.9	2.5	15.0	22.1	24.6	25.2	23.9	19.4	12.2	3.5	-1.7		
	10%	WB	-6.5	-5.5	-0.2	7.6	13.5	17.8	19.4	17.5	12.7	6.9	0.6	-4.4		
		MCDB	-5.9	-4.7	1.2	12.2	20.0	23.3	23.9	22.1	17.2	9.7	1.6	-3.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.1	9.4	9.3	11.1	10.3	9.9	10.6	10.5	7.9	6.3	7.5		
		MCDBR	8.1	7.6	8.8	13.9	15.1	13.4	12.1	13.7	15.3	13.1	6.1	7.3		
		MCWBR	8.0	7.3	7.0	7.7	6.4	5.6	5.2	6.5	7.8	7.6	4.9	7.3		
		MCWBR	7.9	7.5	7.6	13.3	13.1	11.7	10.5	11.8	13.5	11.6	5.9	7.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.322	0.374	0.410	0.407	0.407	0.436	0.412	0.368	0.351	0.321	0.291		
		taud	2.014	1.977	1.911	2.082	2.207	2.277	2.194	2.241	2.318	2.232	2.062	1.961		
		Ebn at Noon	592	715	763	804	835	839	802	798	787	692	545	480		
		Edh at Noon	70	106	144	141	132	125	133	119	96	82	67	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.94	1.97	3.45	4.68	5.44	6.00	5.38	4.52	3.08	1.57	0.77	0.60		
		RadStd	0.08	0.11	0.20	0.32	0.57	0.56	0.37	0.32	0.34	0.18	0.13	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page